AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 32 · STUDENT LESSON

Project Problem and Success Test

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

Name the need. Write the contract. Predict the tests.

Plan a small chooser with reproducible expected behavior before implementation.

Keep one habit: User card → schema and scope → filter/score/tie → acceptance cases → independent plan.

MAIN LESSON · SECTION 1

What must the chooser do before we build it?

> Turn a user need into a reproducible contract and acceptance cases.

Your learning goals

I can define a small application purpose.

I can turn **requirement**s into checkable **acceptance case**s.

I can plan a feasible **pipeline** and human review.

Recall filtering before ranking

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

- Why must a recommendation show no eligible option when every item fails a hard constraint?

Your end goal

By the end, I can plan a recommender with **acceptance test**s before writing its selection code.

▼ Data extension · Preserve invalid durations

What should happen to an invalid duration instead of silently making it zero?

> **My readiness explanation**

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Permissions and Human Review. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from **Evidence** and Sources. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Image and Voice **Pipeline**s. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 32: Project Problem and Success Test

A project is defined by **user need** and measurable success, not feature count.



THINKER MISSION

Write a recommender brief and three **acceptance test**s.

Thinker name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

Readiness for A reproducible data investigation

This week’s end goal

MAIN LESSON · SECTION 2

Plan a small, bounded application

> Name the purpose, decision owner and non-goals.

The20-minute activity-chooser brief

Plan a fictional activity chooser for a teacher-supervised 20-minute session. It will validate a small local dataset, exclude over-limit items, rank eligible items using 2 * fun + learning and break ties alphabetically.

It will explain the result for human review. It will not book, message or collect personal student profiles.

▼ **Data extension · A different accepted-duration question**

Investigate the fictional question: among supplied sessions with valid duration records, what is the mean duration? **Input**s are 10, 0, 20, None and -2 minutes. Our schema accepts exact integers 0 through 60.

The wording deliberately limits the answer to accepted records.

▼ Earlier wording, alternatives and picture transcripts

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



MYSTERY

The learners must use project problem and success test to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

Apply the connected idea: A reproducible data investigation

MAIN LESSON · SECTION 3

Specify fields, validation, score and tie policy

> A short requirements dictionary records settings; it does not implement the chooser.

Predict the printed configuration

```
requirements = {"limit": 20, "fun_weight": 2,
                "learning_weight": 1, "tie": "name ascending"}
print(requirements["limit"])
print(requirements["tie"])
```

Trace the planned pipeline

Follow the evidence

STEP OR CASE	EVIDENCE	MEANING
Input	Local fictional records	Validate schema
Decision	Filter then score then tie rule	Explain selected item
Output	Suggestion or no eligible item	Teacher reviews

Read each row from left to right and connect the evidence to the stated rule.

Fix the schema and testable requirements

Define fields name, minutes, fun and learning. Require a nonempty text name, integer minutes 0 through 60 and integer ratings 0 through 5. Reject duplicate names because the tie rule uses names as identifiers.

Write cases before implementing so **requirement**s do not drift to fit **output**s.

A phrase such as make a smart choice does not say what counts as success.

The chooser plan does: validate the named fields, include durations at most 20 minutes, calculate 2 times fun plus learning, and use alphabetical names to break a tie.

Describe the expected **output** and a no-match result before coding. Another learner should be able to use your plan to predict the same answer without guessing what you intended.

Complete the missing edge-case policies

For this planning contract, integer means an exact integer, excluding Boolean values. Names are unique, case-sensitive identifiers; use simple uppercase A/B/C in supplied tests and ordinary ascending string order. Reject invalid rows with reasons; reject every occurrence of a duplicated name before ranking so input order cannot select a different duplicate. An invalid request is a request error, distinct from a valid request with no eligible items. Record these choices before implementing; a different policy needs its own version and expectations.

Explain the deciding step

- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Data extension · Complete accepted-duration script

```
raw = [10, 0, 20, None, -2]
valid = []
rejected = 0
for value in raw:
    if type(value) is int and 0 <= value <= 60:
        valid.append(value)
    else:
        rejected = rejected + 1
print(len(valid), rejected)
if len(valid) > 0:
    print(sum(valid) / len(valid))
else:
    print("No valid observations")
```

Follow the evidence

STEP OR CASE	EVIDENCE	MEANING
Raw evidence	Five supplied values	Keep unchanged
Validation	Accept 10, 0, 20	Reject unknown and negative
Summary	30 divided by 3	Mean 10 minutes

Read each row from left to right and connect the evidence to the stated rule.

The result is 3 accepted, 2 rejected, and mean 10.0 minutes. A reproducible report states the question, raw **input** version, schema, exclusion reasons, calculation and **scope**.

Someone else should obtain the same result from the same **input**s and rules.

First filter the raw durations using the written schema. Keep 10, 0 and 20; reject None because the duration is unknown and -2 because it is outside the range.

The accepted sum is 30 and the accepted count is three, so the accepted-record mean is 10 minutes. Dividing by all five supplied records would treat the two rejected durations as if they contributed known zeros.

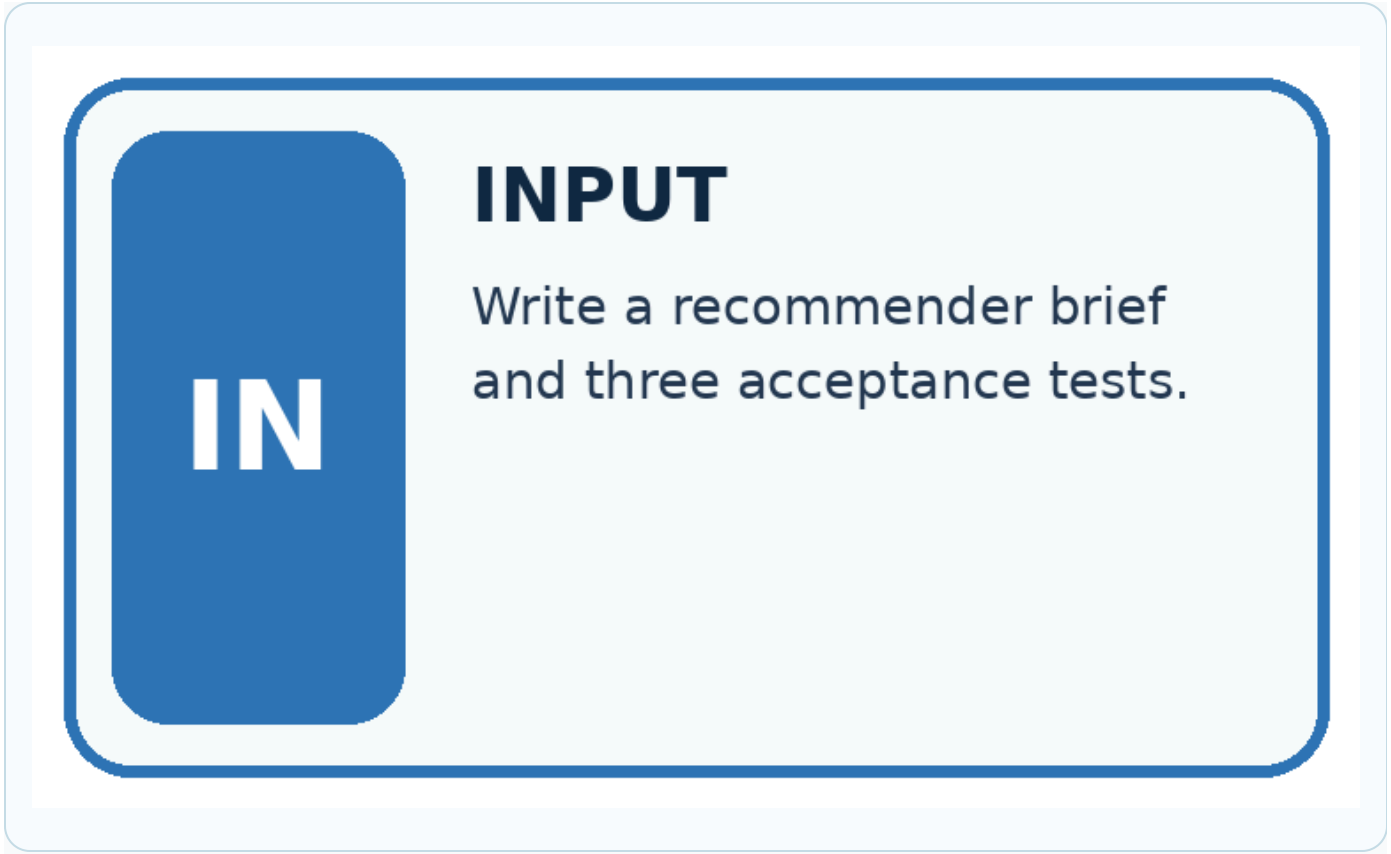
State three accepted out of five beside the mean.

▼ Earlier wording, alternatives and picture transcripts

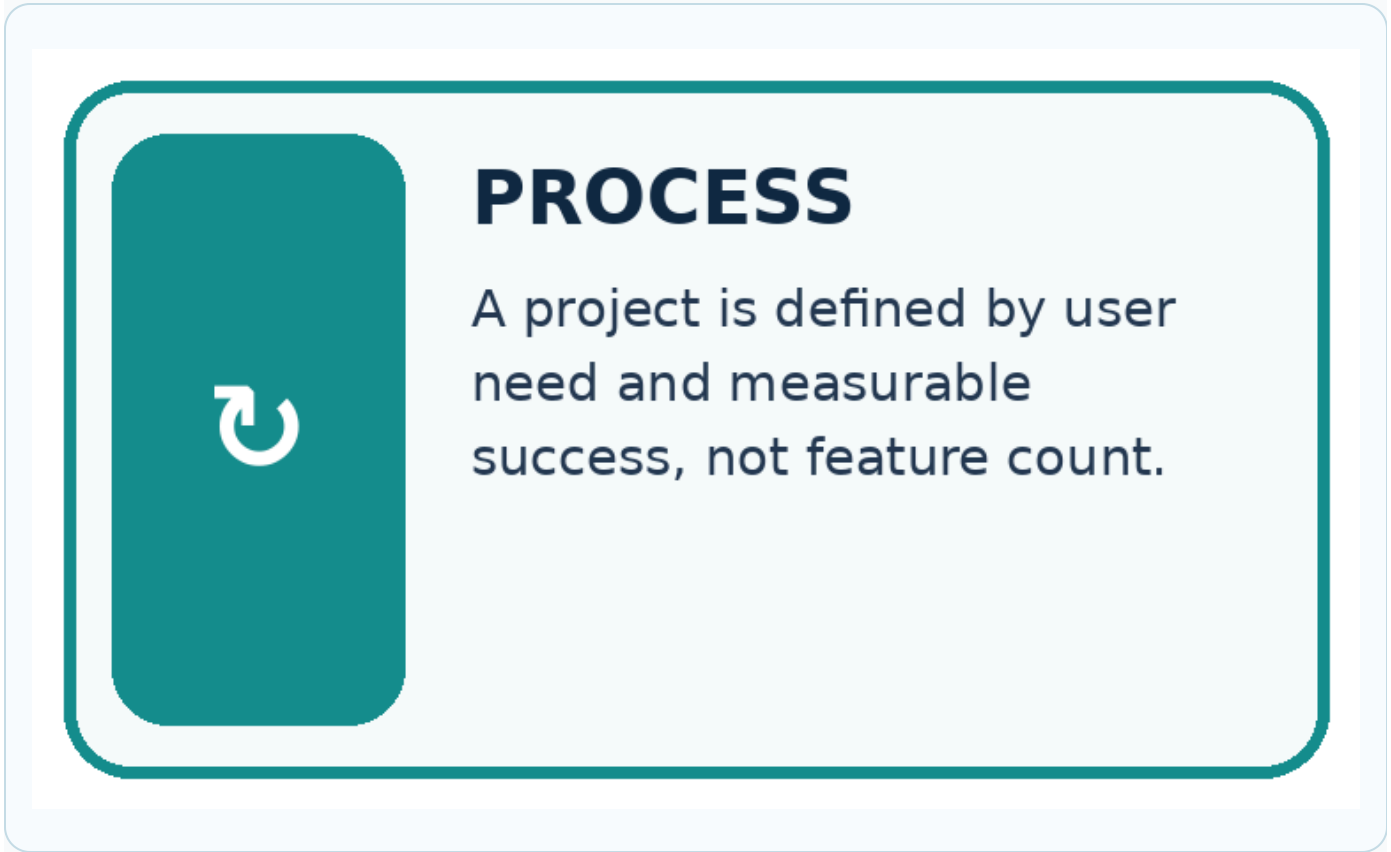
VISUAL **MECHANISM**

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.


IN
INPUT
Write a recommender brief and three acceptance tests.

↓


↻
PROCESS
A project is defined by user need and measurable success, not feature count.

↓



Which stage could fail even when the other two are correct? Explain:

> **Write or draw your reasoning and evidence.**

A closer explanation

Turn a wish into a checkable plan

Apply the connected idea: A reproducible data investigation

Keep the denominator with the evidence

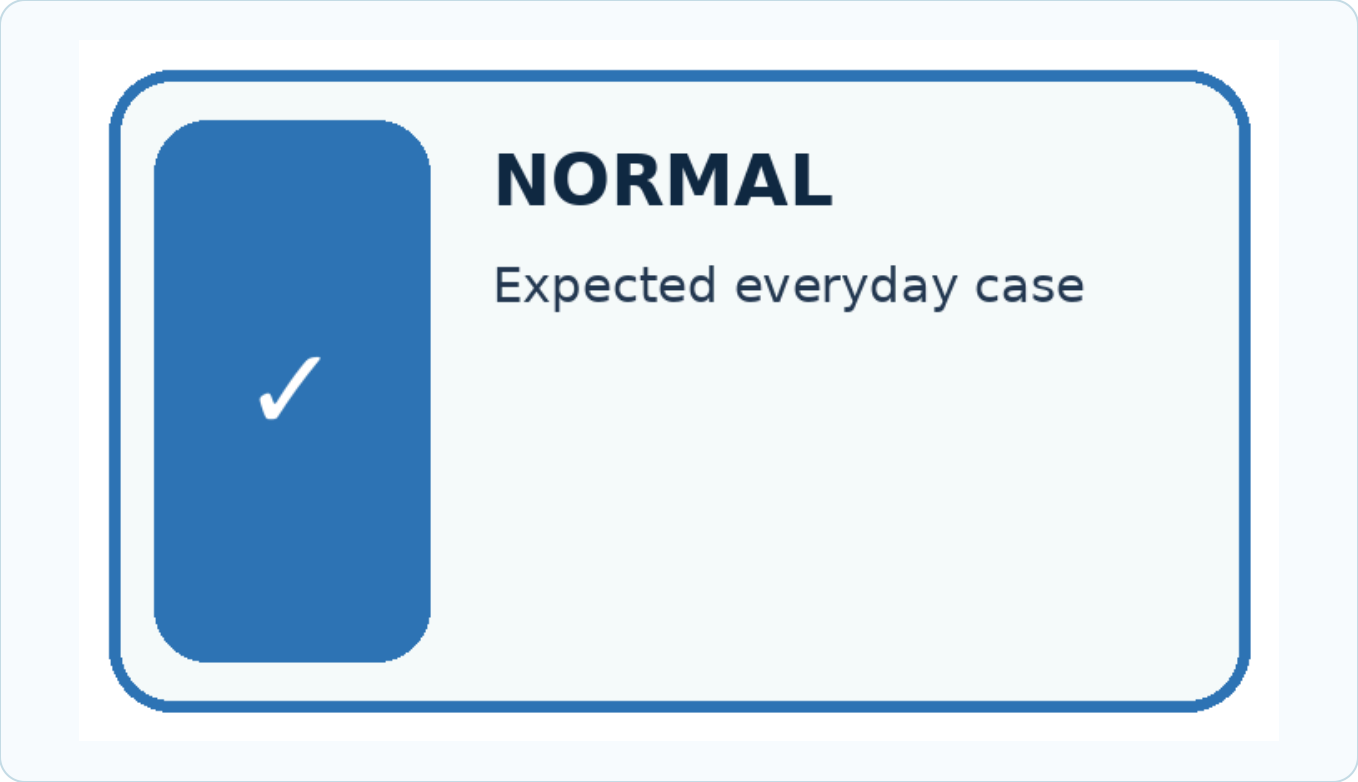
Connect the picture and the explanation

MAIN LESSON · SECTION 4

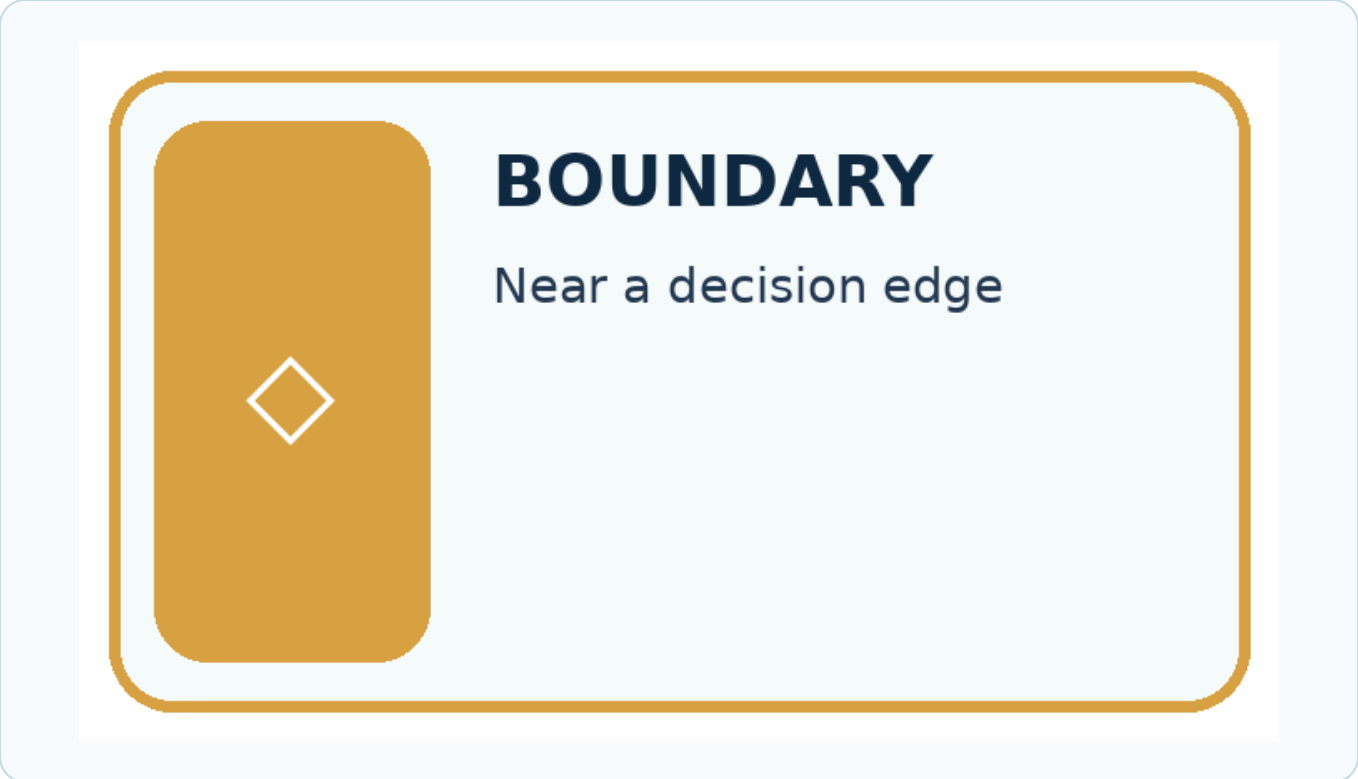
Compare testable requirements with vague promises

> Write expected behavior before seeing program output.

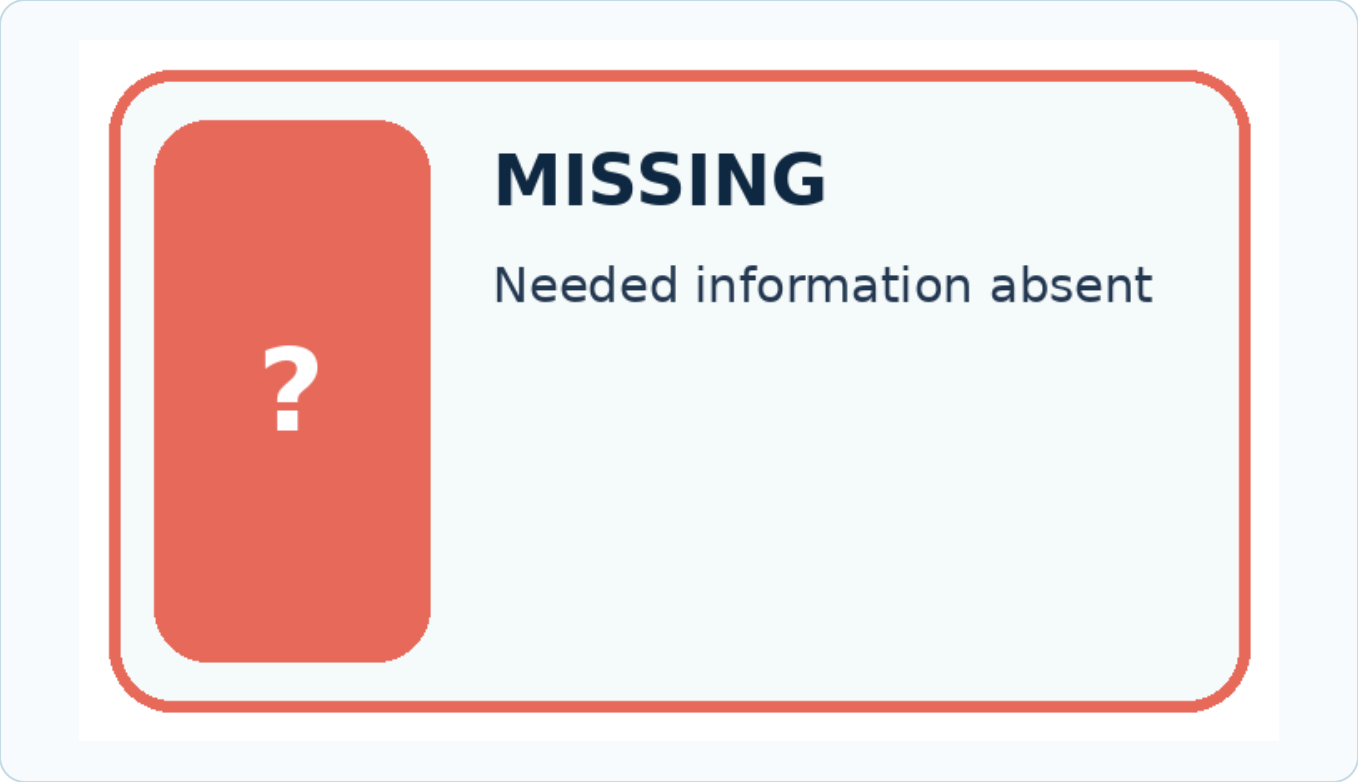
Four pictures with a purpose



Normal case: predict the result and deciding evidence.



Boundary case: identify the exact limit.



Missing information: name what must be clarified.



Transfer case: explain which procedure still applies.

Choose a revealing case

- Which case is most likely to reveal a hidden assumption? Why?

> **Write or draw your reasoning and evidence.**

Predict the supplied changed case

- Change: Reduce the limit from 8 to 5.

▼ Check after predicting

Prediction: Return no eligible item.

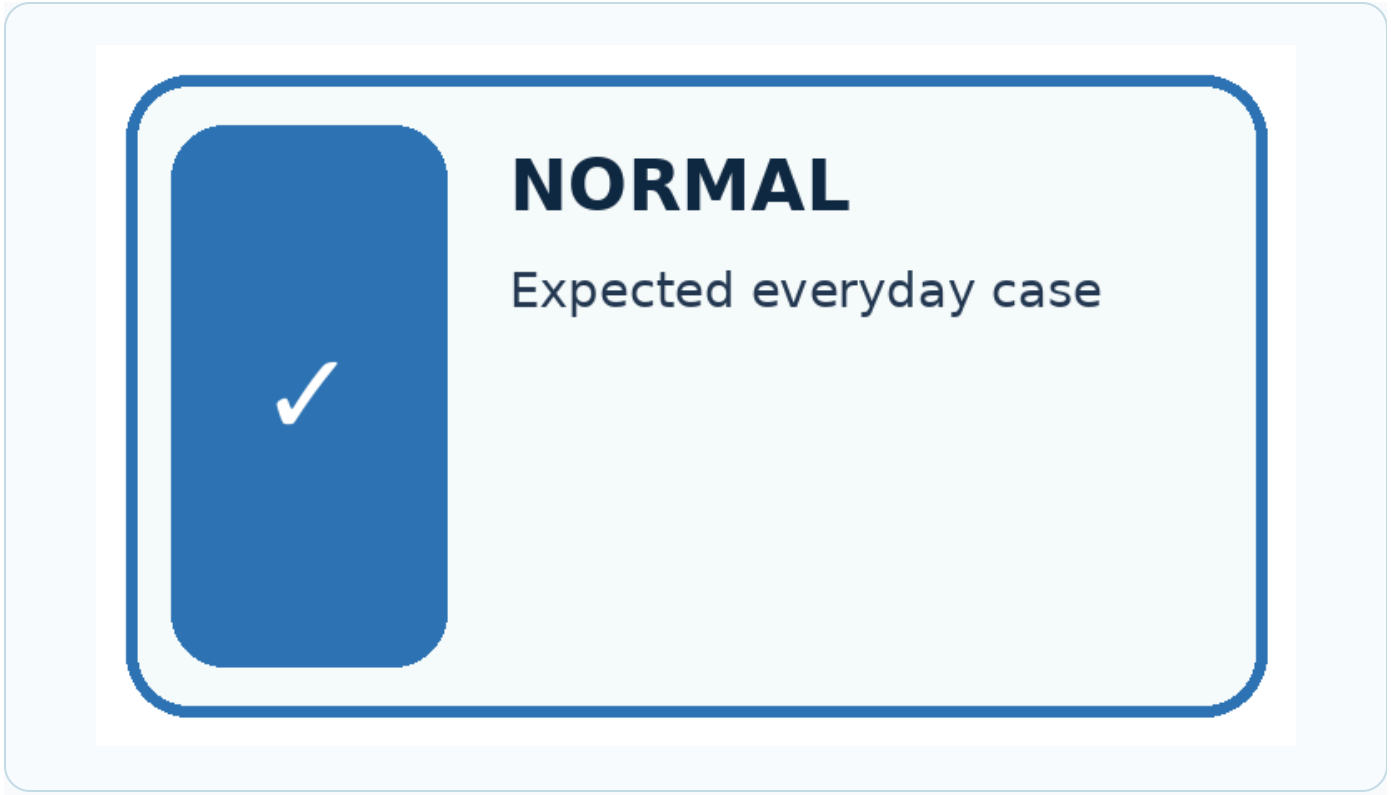
Reason: All three valid rows take more than five minutes.

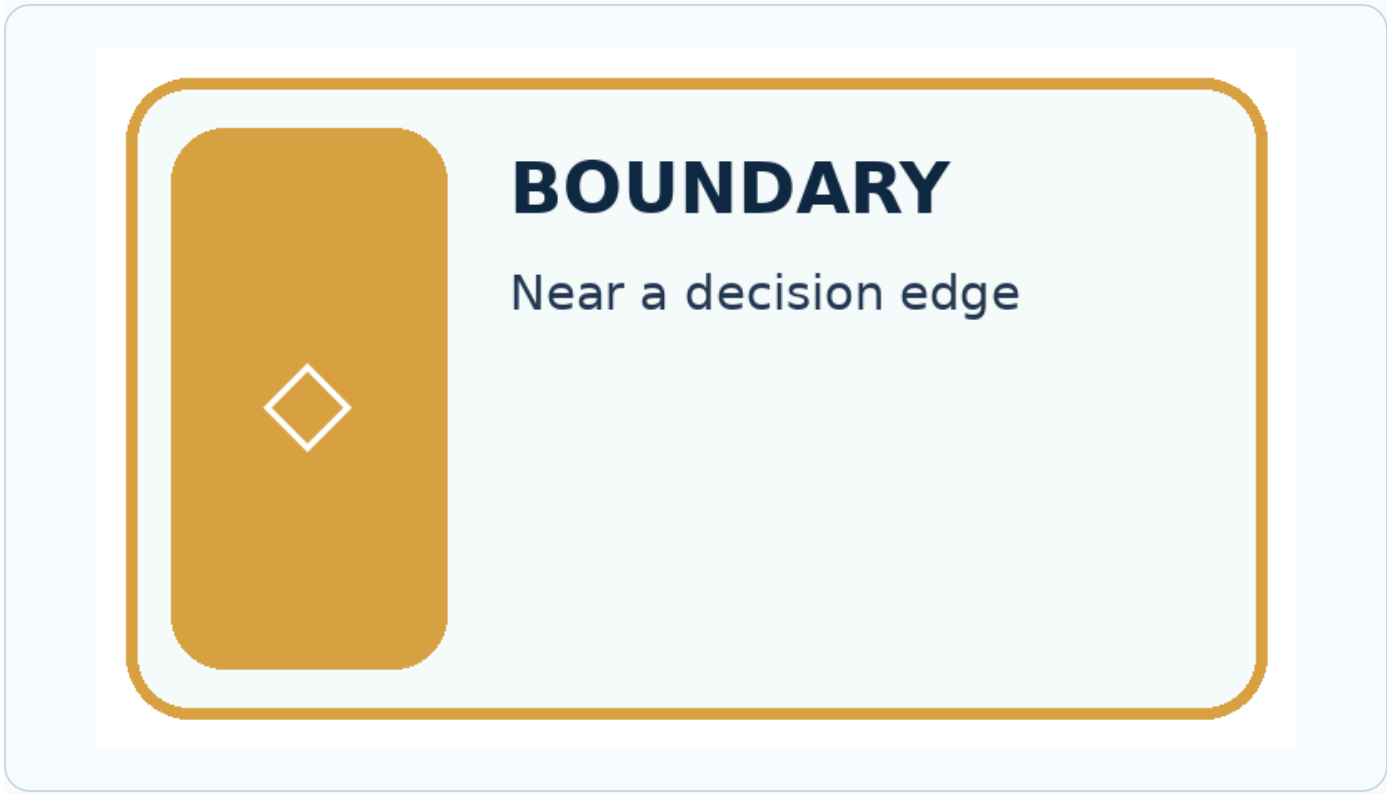
▼ Earlier wording, alternatives and picture transcripts

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.





NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Write main acceptance cases, then inspect a separate pictured chooser

> The activity score2fun+learning differs from the pictured interest-only score.

Read concrete main-contract acceptance cases

Main20-minute chooser: acceptance input	Expected behavior from the contract
A: minutes20,fun3,learning2; B:21,5,5	A eligible score8; B excluded despite score15
A:10,3,2; B:12,2,4; reverse row order too	Both score8; A selected in either order
A:21,3,2 only	No eligible option; do not silently relax the limit
Request limit True	Reject invalid request; do not treat True as1
Two rows both named A	Reject both duplicate-name rows with reasons

Write the boundary and no-match cases yourself

Write an acceptance case for the duration boundary and one for no eligible option.

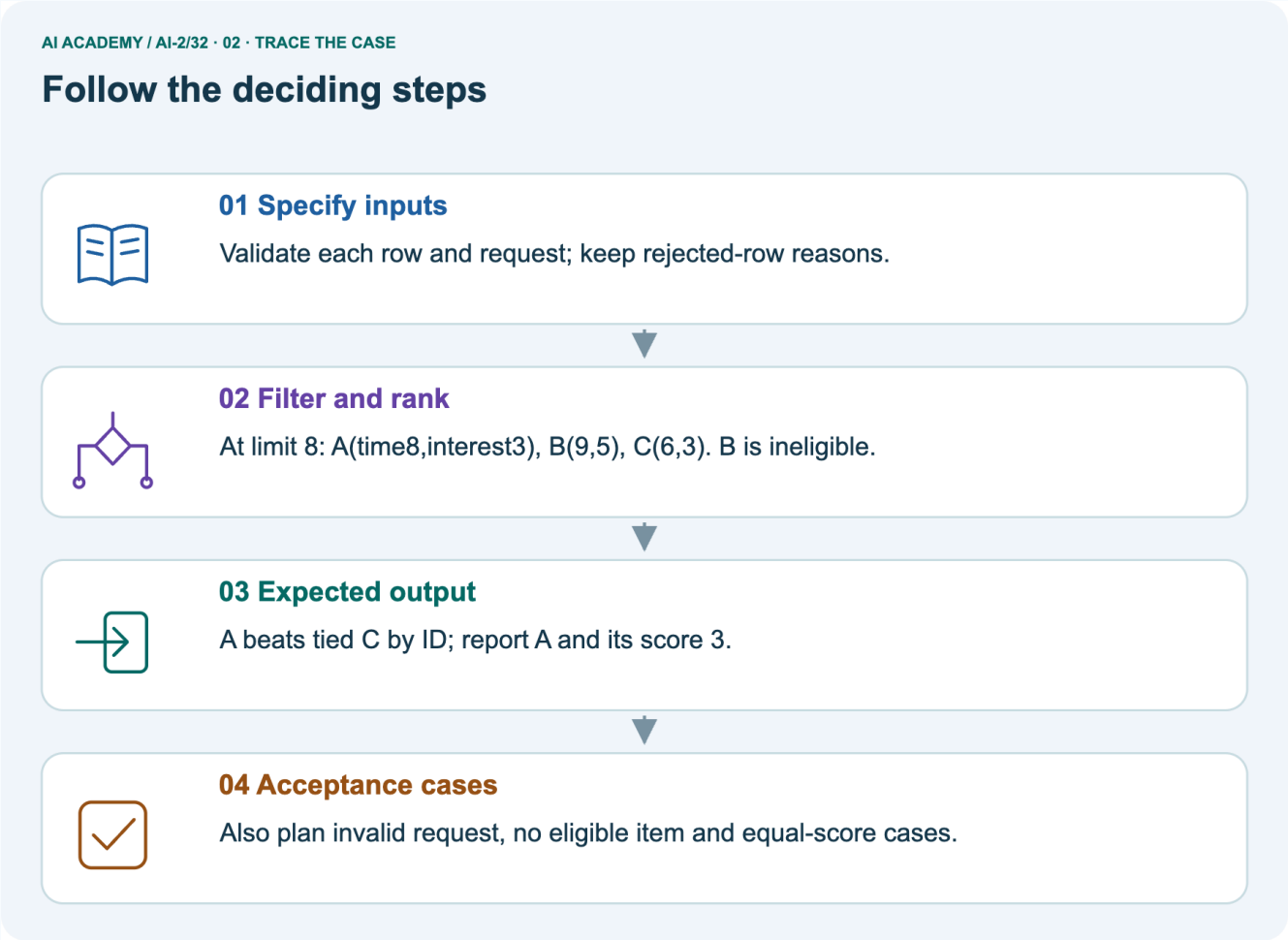
Start the separate reading-break contract

A fictional reading-break chooser needs a transparent contract.

Items have id text, minutes integer 0–30, interest integer 0–5. Request limit is integer 0–30.

Eligible if minutes<=limit; score=interest; highest wins; ties choose alphabetically smaller ID; invalid rows excluded and counted.

Trace its limit8 and ID tie



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Specify inputs	Validate each row and request; keep rejected-row reasons.
2. Filter and rank	At limit 8: A(time8,interest3), B(9,5), C(6,3). B is ineligible.
3. Expected output	A beats tied C by ID; report A and its score 3.
4. Acceptance cases	Also plan invalid request, no eligible item and equal-score cases.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

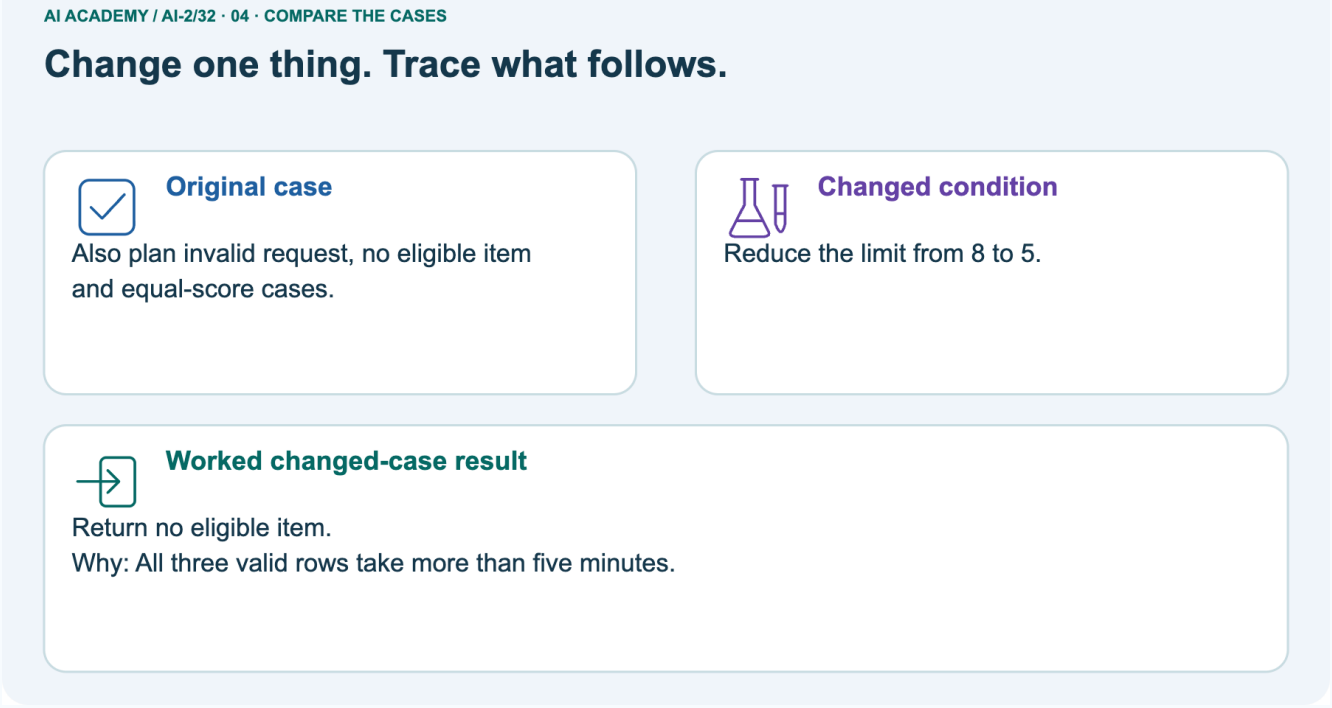
Planning fixes what success means before code returns an answer. The duration boundary includes equality, and the tie policy prevents accidental list-order dependence.

A complete **output** may include why other rows were excluded. Test expectations should come from the contract rather than copying the first program result.

Predict the limit5 change

- Change one thing: Reduce the limit from 8 to 5. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain scope and equality

- The supplied interest scores are fictional; the specification does not establish genuine learner preferences.
- Read the labelled picture: Write an equality test and explain which **requirement** it checks.

▼ Data extension · Add30 to the accepted-duration list

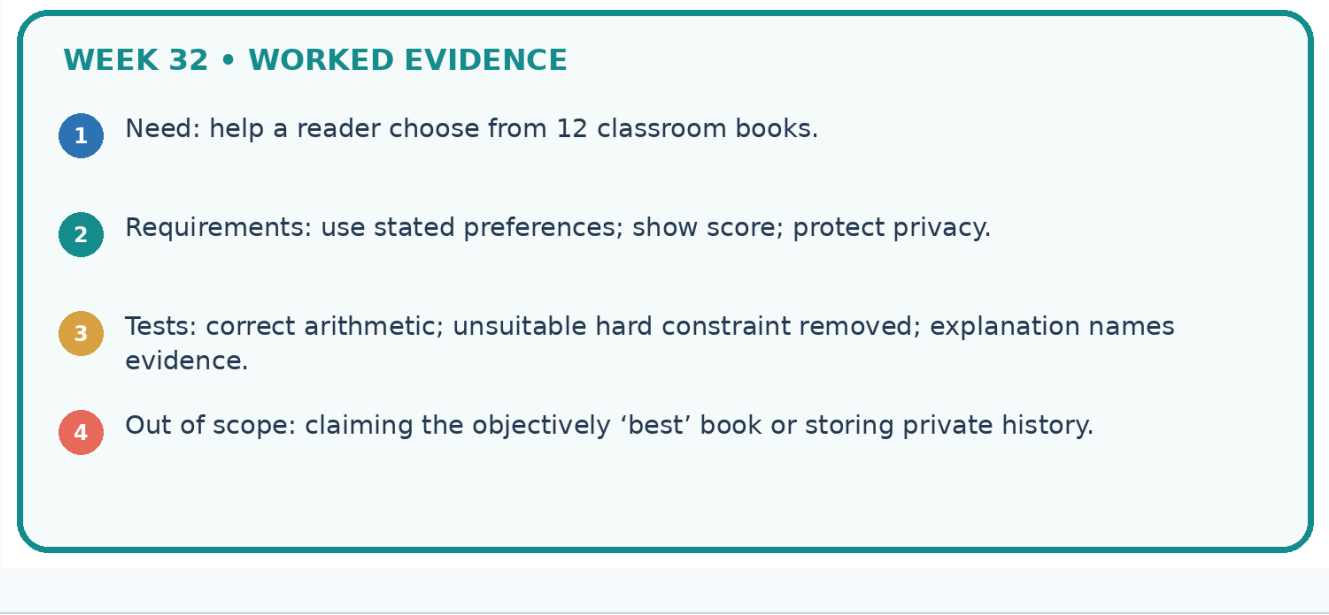
> **If another valid value 30 is added, calculate the new accepted count and mean.**

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. Need: help a reader choose from 12 classroom books.

2. **Requirement**s: use stated preferences; show score; protect privacy.

3. Tests: correct arithmetic; unsuitable hard constraint removed; explanation names **evidence**.

4. Out of **scope**: claiming the objectively ‘best’ book or storing private history.

Recalculate, retrace, or restate the reasoning in your own representation:

> **Write or draw your reasoning and evidence.**

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

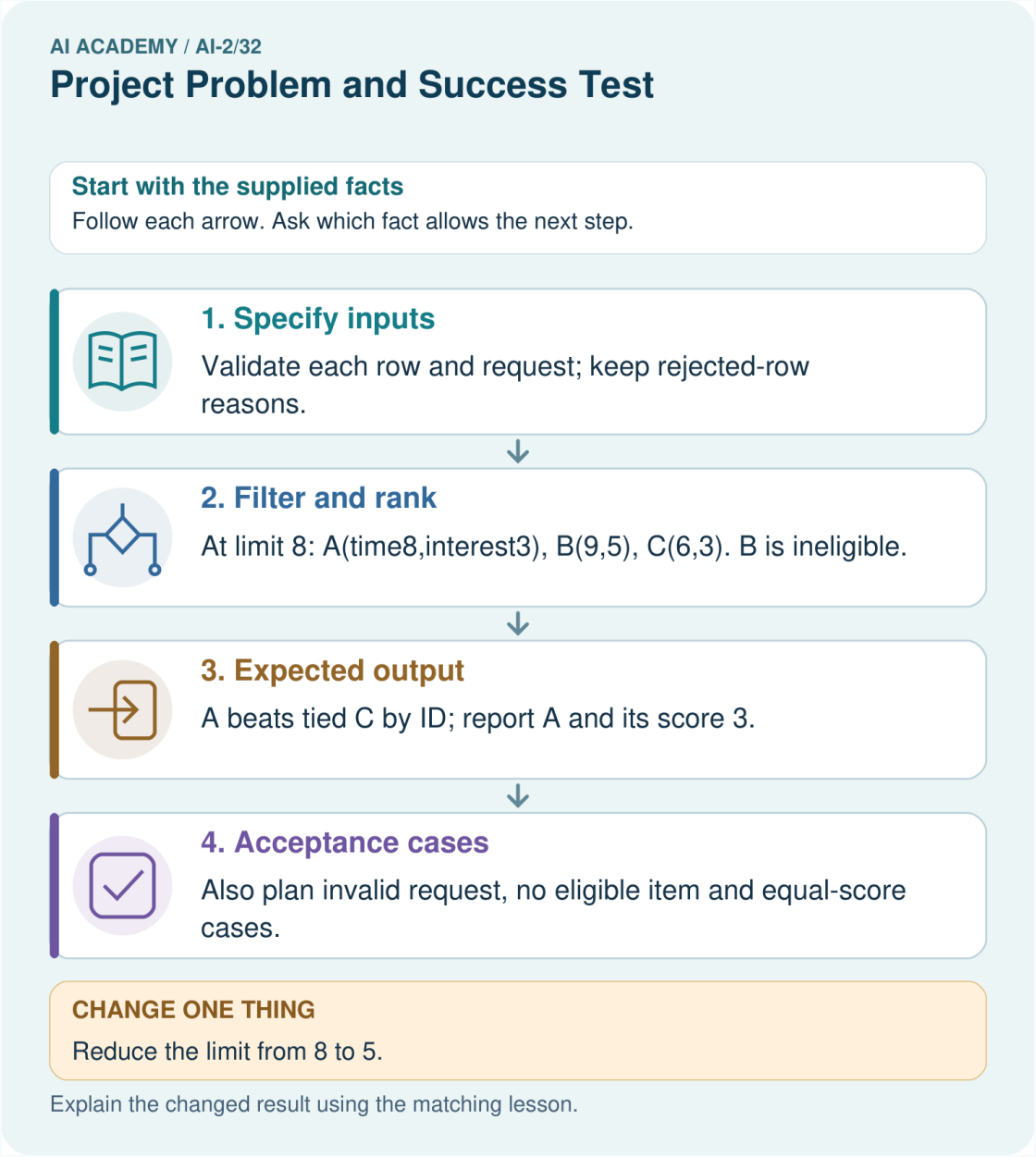
Apply the connected idea: A reproducible data investigation

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Project Problem and Success Test

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Replace a vague success claim with a checkable one

> A requirement must let another learner predict the expected output.

Repair the best-activity claim

Claim to repair: A prototype returned A once, so it always chooses the best activity. A testable claim is narrower: under the declared schema,20-minute limit,2fun+learning score and name tie policy, this input should return the expected eligible item. Add cases for equality, ties and order, duplicates, invalid requests and no match. A correct trace does not establish real preferences, learning benefit or every future case.

Design one case that could reveal a failure

Name two capabilities excluded from this prototype and explain why.

▼ **Earlier wording, alternatives and picture transcripts**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: A project is better when it has more features. Strong move: locate evidence, change one cause, and retest.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

> **Write or draw your reasoning and evidence.**

A closer explanation

The **scope** is a rule-based **prototype** using fictional ratings. It does not demonstrate personalized learning impact or a trained AI model. A clear **non-goal** protects students from building an untestable or overly broad project.

Apply the connected idea: A reproducible data investigation

Missing or invalid durations might belong to unusually long sessions. The accepted mean may therefore differ from the full group's mean. Do not make a causal claim from this descriptive **investigation**.

> **A report gives only mean 9. What additional evidence lets a peer reproduce it?**

MAIN LESSON · SECTION 8

Practise writing an acceptance case with less help

> Use actual input fields and an expected outcome.

Replace vague scope with an explicit contract

A **requirement** such as make a smart choice cannot be tested precisely. Replace it with a stated score, limit, tie rule and **output** explanation.

If real feedback later changes the policy, record a versioned change rather than rewriting previous **evidence**.

Complete main W2

Practise with less help: Write an acceptance case

Use workbook W2 for the connected example “Planning the application”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Repair a one-run overclaim

- Use the worked case for Planning the application. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Check your reasoning

Explain the separate pictured equality test

- Write an equality test and explain which **requirement** it checks.

Explain the picture

> **My explanation using a pictured step and the given facts**

Keep independent planning separate

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Data extension · Audit the accepted denominator

Dividing the accepted total 30 by all five records gives 6 and answers no clear observed-duration question.

Use the accepted count for this mean, disclose exclusions, and never describe it as the average for every supplied session.

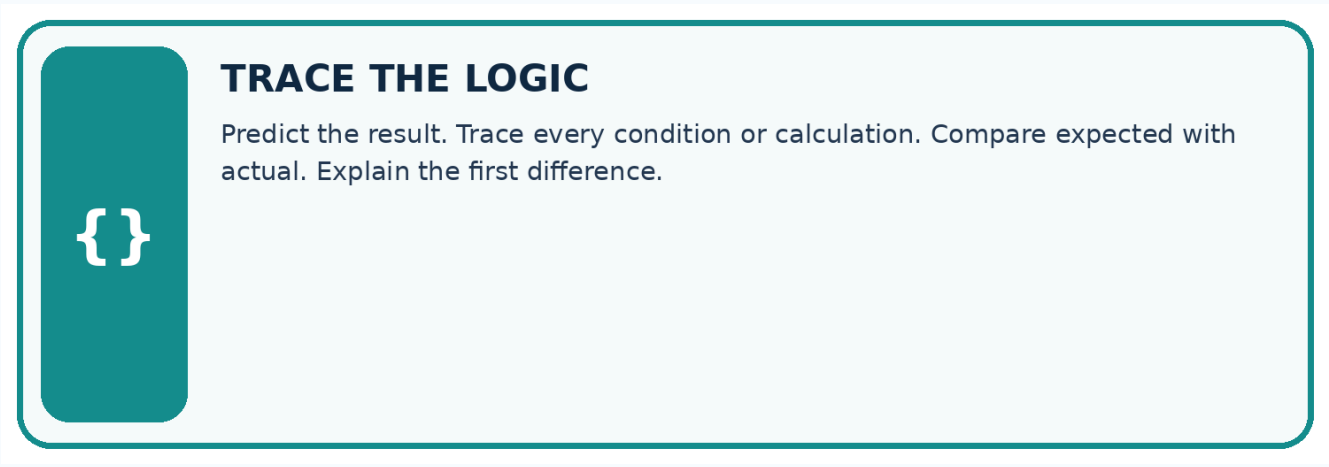
Use its named workbook W2. Identify the given **input**s, try the middle steps with less help, then explain your trace. Check this claim before group discussion: A successful run proves the conclusion

▼ Earlier wording, alternatives and picture transcripts

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



SET **evidence**_record PREDICT expected_result RUN one normal case and one boundary case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, **limitation**, and next safe step

Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Apply the connected idea: A reproducible data investigation

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Practise the connected topic: A reproducible data investigation

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MAIN LESSON · SECTION 9

Produce a complete specification and acceptance table

> The investigation supplies its user need and inputs; implementation comes later.

The fictional user card

Fictional user card: a teacher needs one activity suggestion for a20-minute supervised session. Provided records are A(nameA,minutes20,fun3,learning2), B(B,21,5,5), C(C,12,2,4). Use exact integers and the main2fun+learning contract; no personal profiles. The teacher retains the decision. Non-goals include booking, messages and claims of proven learning benefit.

Plan, predict and peer-check reproducibility

1. Planning investigation: use the supplied user card rather than collecting real information. Write a one-sentence need, the full schema, validation and duplicate policy, hard limit, score, tie rule, expected output and human-review boundary. Before implementing, predict eligible rows and scores for the supplied records, then author concrete equality, tie/order and no-match cases. Add one invalid row and one invalid request; say which error each should produce. Exchange plans with a partner who checks whether the same expectations follow without guessing. Keep plan-only, paper trace and future execution results labelled separately. For the separate duration extension, preserve the raw list and rejection reasons before calculating its accepted-record mean. Reserve the15-minute chooser brief for independent work.

Keep expected and actual evidence separate

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____

> **Change or keep?:** _____

> **Test / input:** fresh transfer
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

> **Write or draw your reasoning and evidence.**

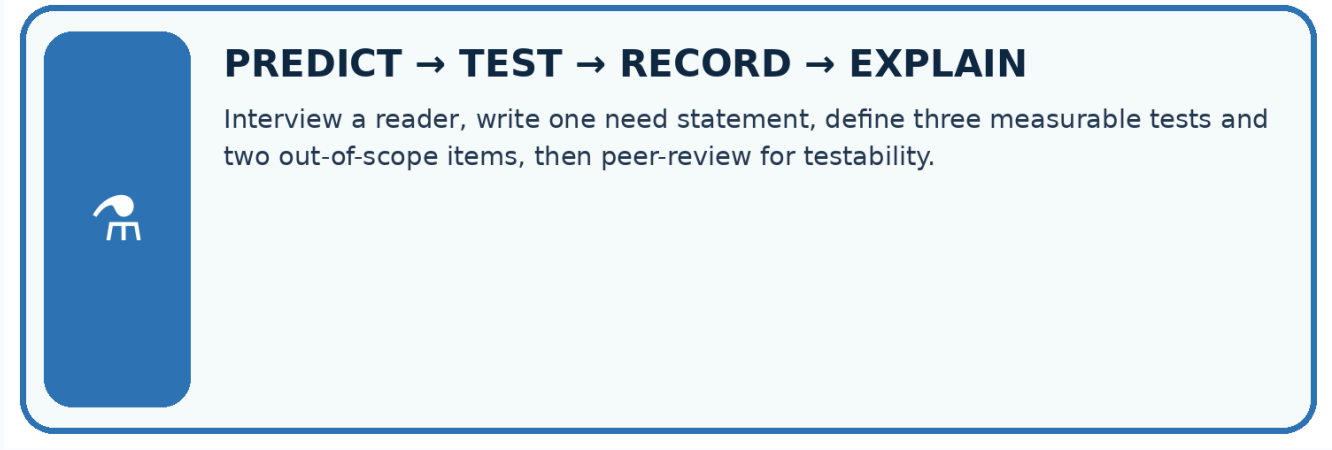
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Use the supplied fictional teacher/session card and records. Write measurable requirements, concrete acceptance inputs and expected outcomes before implementation. Peer-review the plan without interviewing or collecting personal reader data.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: plan the fresh15-minute club chooser

> Write the contract and expected outcomes before any implementation.

Independent brief and evidence scope

Plan a 15-minute fictional club chooser. Specify schema, score, duration limit, tie rule, invalid-data behavior, **output** and three **acceptance case**s.

Use invented records and no live service. Either retain the main schema/score with the requested15-minute limit or state and justify a complete alternative explicitly. Your three acceptance cases must contain actual input records, predicted outputs and reasons, not just names such as boundary or tie. Include the human-review boundary and distinguish a specification from executed evidence.

Your first attempt and help record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Plan independently

Week 32 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Define a testable goal for a book recommender serving readers who want short space books. Include a concrete success condition.

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Freeze the brief before building so later choices can be traced to **requirement**s.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

Specify three **acceptance test**s for a recommender: a normal match, a tied score and no qualifying book. Include expected behaviour before implementation.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Freeze a specification the next build can follow

> Keep requirements and acceptance expectations alongside the project version.

This week’s concrete contribution

Version the recommender brief before building: purpose, allowed inputs, schema, hard constraints, score, stable tie rule, invalid-request response, rejected-row reasons, no-match output and human decision owner. Map every requirement to an acceptance case with expected results. Treat activity chooser examples as design practice; retain the actual book catalogue schema and evidence requirements in the book project. Do not silently exchange interest-only and2fun+learning rules.

Retain the Transparent Book Recommender’s schema, constraints, score, tie rule and acceptance table as its build specification.

Keep the checkpoint and human choice

THIS WEEK’S CHECKPOINT Freeze the brief before building so later choices can be traced to **requirement**s.

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked candidates → explanation → test record → **human decision**

What will you add, change, or verify in the project this week?

> **Write or draw your reasoning and evidence.**

What should remain under **human control**?

Write or draw your reasoning and evidence.

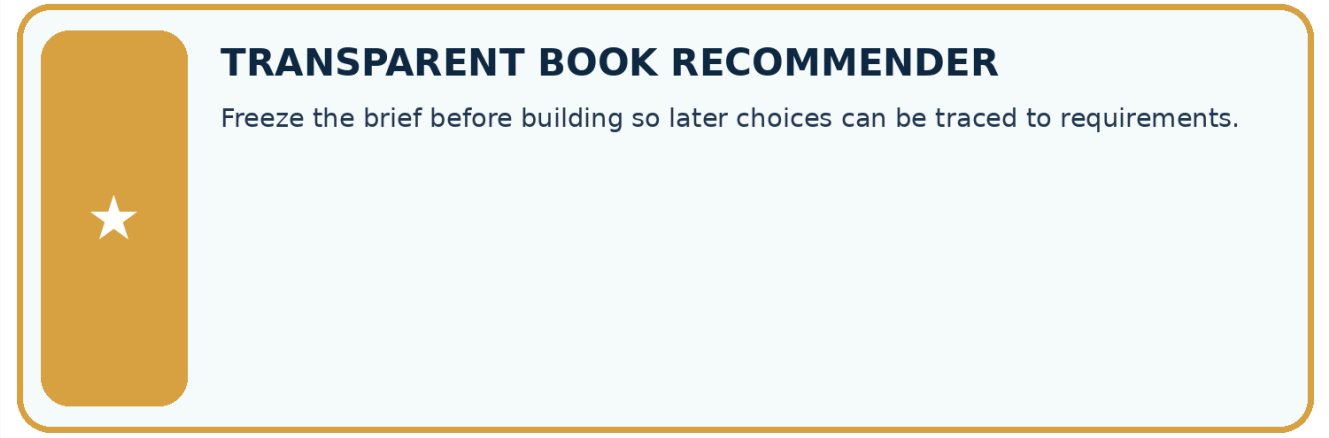
RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits.
Never confuse a ranking with a fact about a person.

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



Evidence for your project

Finish: show how each requirement can be checked

> A precise plan lets another learner reproduce the expected decision.

Explain without notes

- Close your notes. Explain this goal in your own words: Define a small application purpose

> **My deciding step, evidence and one remaining question**

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Words for the main lesson and extension**

REFERENCE · SECTION 5

Reference · Words for the main lesson and extension

> Use each word with a concrete record or calculation.

Core words

Word	My own example	Not this / boundary
user need	The problem or desired outcome the system is meant to address for a person.	
requirement	A behaviour or condition the proposed system must satisfy.	
acceptance test	A predefined check that a product satisfies a stated requirement.	
scope	The tasks, users, information and actions included within an agreement or design.	

Main explanation terms

Word	Meaning
Acceptance case	An input and expected behavior used to judge a requirement

Word	Meaning
Non-goal	Something intentionally outside project scope
Prototype	A limited implementation for testing a design

Extension terms

Term	Meaning
Investigation	A documented process for answering a question with evidence
Pipeline	An ordered sequence of processing steps
Audit trail	A record of inputs, decisions and changes
Limitation	A reason a conclusion has restricted scope

▼ Earlier wording, alternatives and picture transcripts

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.


1
USER NEED
specific problem for a specific person


2
REQUIREMENT
behavior the solution must provide





Words used in the closer explanation

Terms for A reproducible data investigation

▼ **Extra challenge · Choose a stronger explanation**

REFERENCE · SECTION 10

Extra challenge · Choose a stronger explanation

> Choose the next useful challenge rather than repeating the ladder.

Thinking ladder

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.
- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.
- My strongest **evidence** of independent understanding:

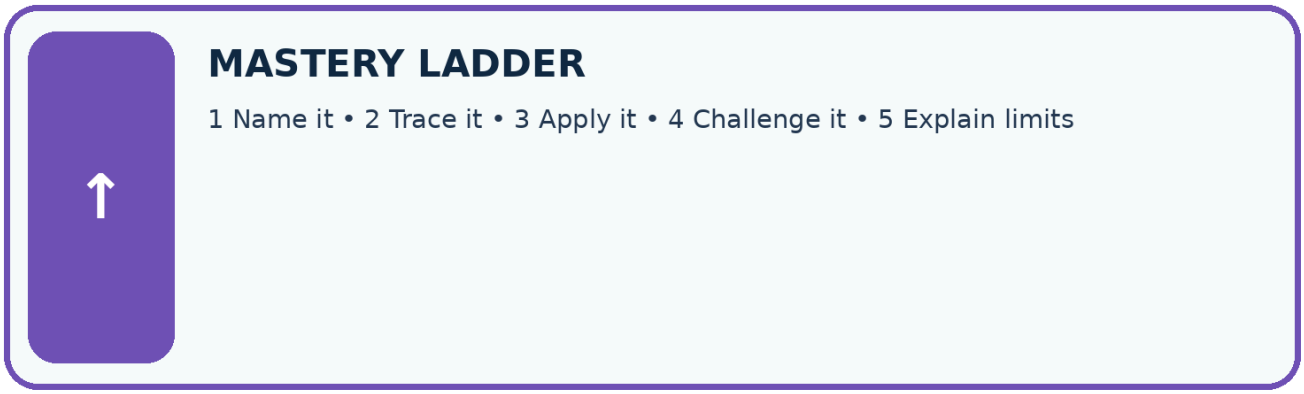
> **Write or draw your reasoning and evidence.**

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



▼ Reference · Check your explanation

REFERENCE · SECTION 12

Reference · Check your explanation

> Use after practice; Plan a new chooser independently is the selected independent task.

Evidence and limits

- 1. My fresh example, trace, calculation, or design:
- 2. My **evidence** that it works (include a boundary or missing-data case):
- 3. One **limitation**, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Write or draw your reasoning and evidence.

Write or draw your reasoning and evidence.

Self-check

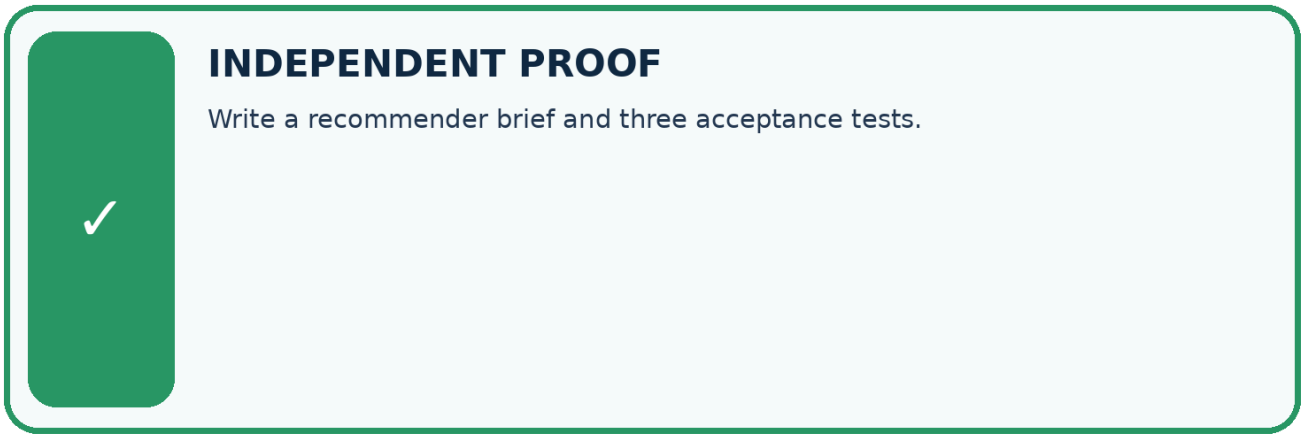
SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a **limitation** ☐ I can answer ‘why?’

▼ Earlier wording, alternatives and picture transcripts

INDEPENDENT MASTERY PROOF

Checklist for this week’s explanation

Use after practice. Plan a new chooser independently is the selected independent task.



Require full input fields, explicit decisions and predicted acceptance outcomes, with no claim of execution.

▼ **Extra practice · The earlier complete case**

REFERENCE · SECTION 13

Extra practice · The earlier complete case

> Try the question before opening the worked explanation.

Question and given facts

A project recommends from ten approved books using genre and length.

Define **acceptance test**s.

Your explanation

- Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

▼ **Check after trying**

Verify book existence, hard constraints, correct scoring explanations, tie handling and no-match behaviour on predefined requests.

Evidence or stage	Value or result
Catalogue	Suggestion must be one of ten approved books
Constraints	Genre and length respected
Scoring	Explanation matches the calculation
Boundaries	Handle ties and no matches

▼ Earlier wording, alternatives and picture transcripts

Week 32 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Turn a project goal into acceptance tests

A project recommends from ten approved books using genre and length.

Define **acceptance test**s.

Verify book existence, hard constraints, correct scoring explanations, tie handling and no-match behaviour on predefined requests.

▼ Reference · Earlier checklist

Reference · Earlier checklist

> Use when it helps inspect the evidence.

Checklist and first prediction

By the end, I can	How I will show it
Define a small application purpose	Explain the deciding step, show a trace or test, and state one limit.
Turn requirements into checkable acceptance cases	Explain the deciding step, show a trace or test, and state one limit.
Plan a feasible pipeline and human review	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: A fictional reading-break chooser needs a transparent contract.

▼ Extra practice · Pictures, feedback and repair

REFERENCE · SECTION 17

Extra practice · Pictures, feedback and repair

> Try before opening feedback; keep the selected independent task separate.

First supported question

- A fictional reading-break chooser needs a transparent contract.
- Which labelled step matches this detail? Also plan invalid request, no eligible item and equal-score cases.
- 1. Filter and rank
- 2. Specify **input**s
- 3. **Acceptance case**s
- 4. Expected **output**
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after trying**

After your attempt, compare with the supported explanation: **Acceptance case**s Also plan invalid request, no eligible item and equal-score cases. Planning fixes what success means before code returns an answer.

The duration boundary includes equality, and the tie policy prevents accidental list-order dependence. A complete **output** may include why other rows were excluded.

Test expectations should come from the contract rather than copying the first program result.

Changed-case question

- Changed-case practice: Reduce the limit from 8 to 5. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. Return no eligible item.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check changed case**

After your attempt, compare with the supported explanation: Return no eligible item. All three valid rows take more than five minutes.

Repair and keep evidence

Review Permissions and Human Review (ai-2/31). Goal: Explain the **mechanism** and terminology of least privilege and tool **scope**s. Short practice: Which proposal is allowed, and why is an authorised object alone insufficient?.

Return to this week and explain what you changed.

Review **Evidence** and Sources (ai-2/29). Goal: Explain the **mechanism** and terminology of claim support and citation evaluation.

Short practice: Why is the original answer not fully supported despite having citations on every claim?. Return to this week and explain what you changed.

Review Image and Voice **Pipeline**s (ai-2/26). Goal: Explain the **mechanism** and terminology of speech **pipeline**s and multimodal **evidence**.

Short practice: Which stage first fails, and why should the faulty transcript not be silently overwritten?. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

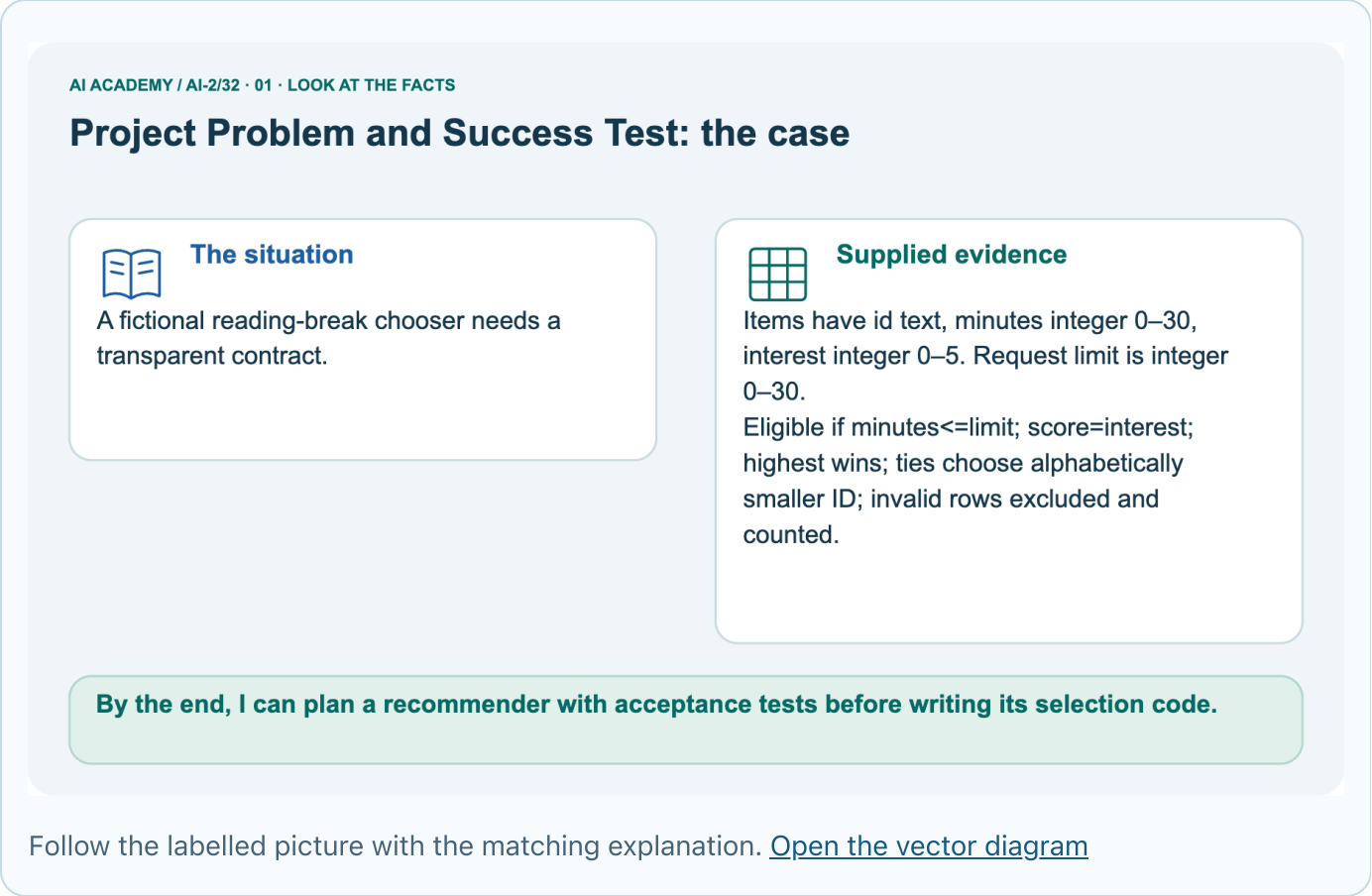
▼ **Reference · The six-picture journey**

REFERENCE · SECTION 18

Reference · The six-picture journey

> Revisit the picture that serves the current question.

01. Project Problem and Success Test: the case



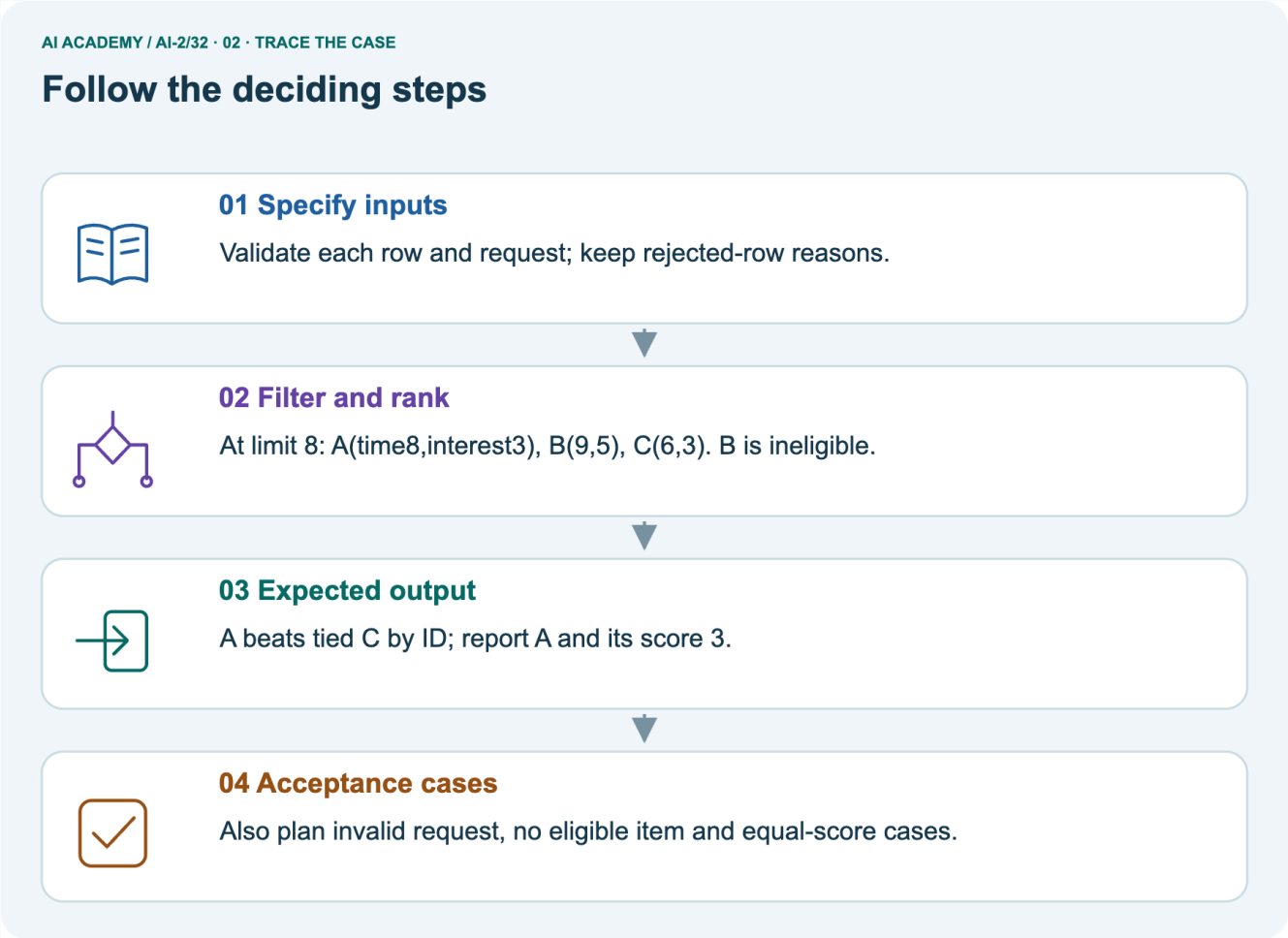
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional reading-break chooser needs a transparent contract. Supplied facts: Items have id text, minutes integer 0–30, interest integer 0–5. Request limit is integer 0–30.

Eligible if minutes<=limit; score=interest; highest wins; ties choose alphabetically smaller ID; invalid rows excluded and counted.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



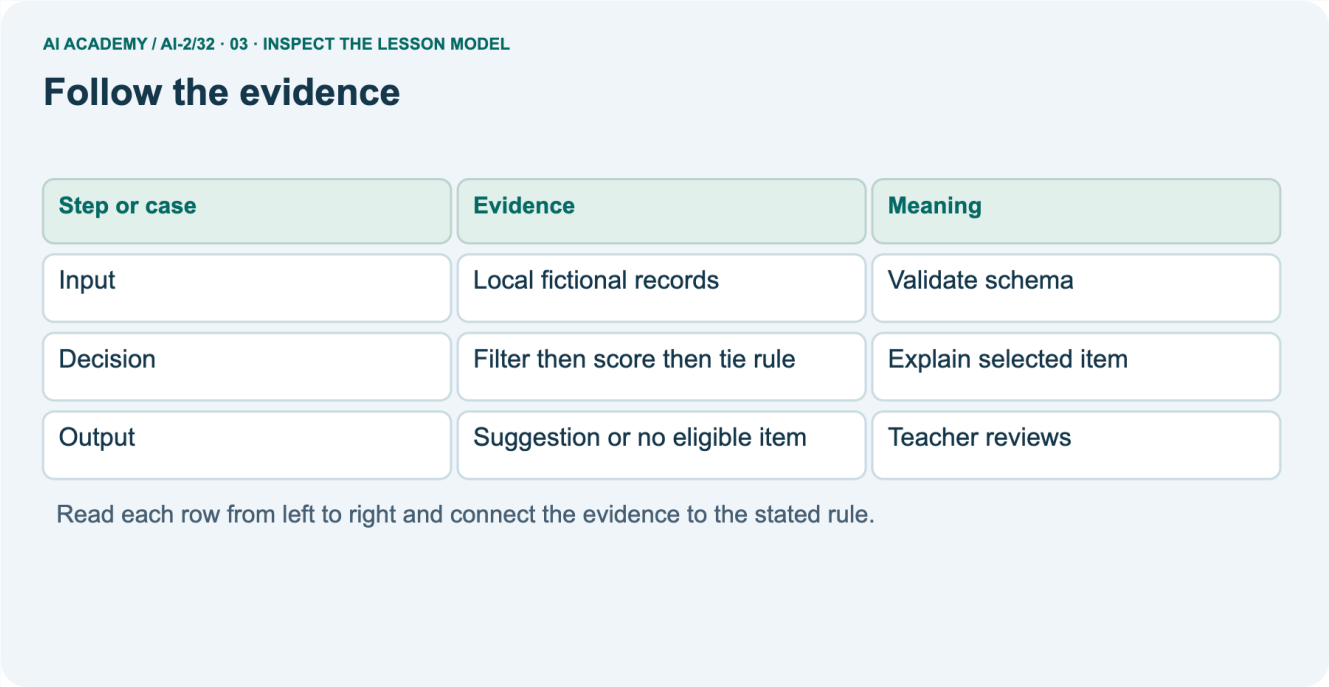
Follow the labelled picture with the matching explanation. Open the vector diagram

Specify **input**s: Validate each row and request; keep rejected-row reasons. → Filter and rank: At limit 8: A(time8,interest3), B(9,5), C(6,3).

B is ineligible. → Expected **output**: A beats tied C by ID; report A and its score 3. → **Acceptance case**s: Also plan invalid request, no eligible item and equal-score cases.

Point to each numbered stage and explain which recorded **input** it uses.

03. Follow the evidence

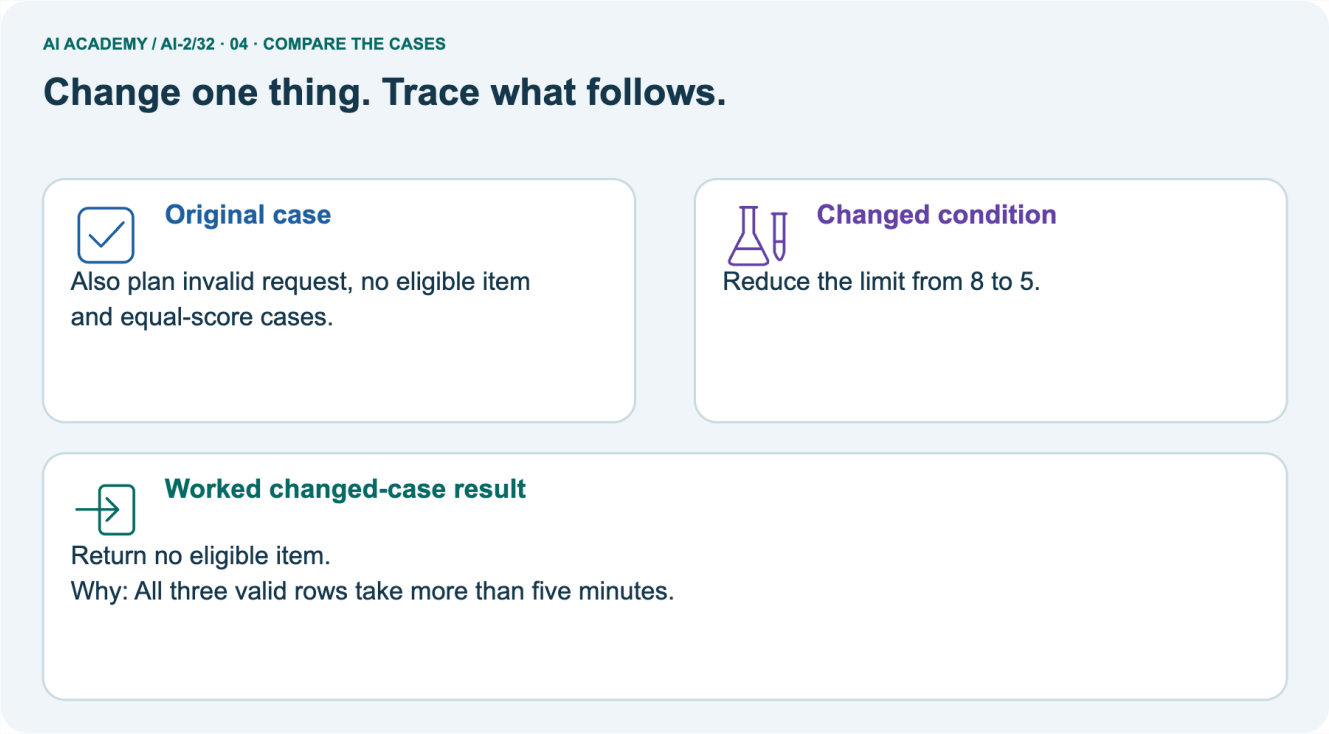


Follow the labelled picture with the matching explanation. Open the vector diagram

Follow the **evidence**. **Input** | Local fictional records | Validate schema. Decision | Filter then score then tie rule | Explain selected item. **Output** | Suggestion or no eligible item | Teacher reviews

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



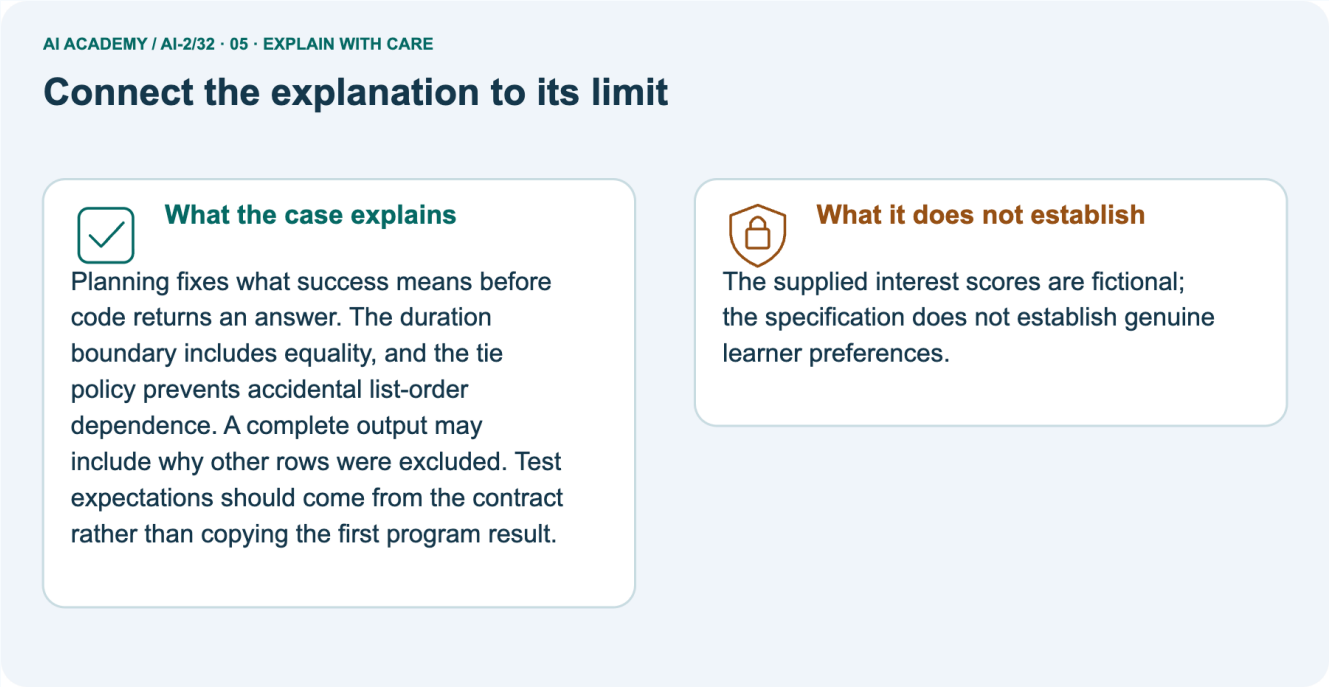
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Also plan invalid request, no eligible item and equal-score cases. Changed condition: Reduce the limit from 8 to 5. Expected result: Return no eligible item.

Reason: All three valid rows take more than five minutes.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

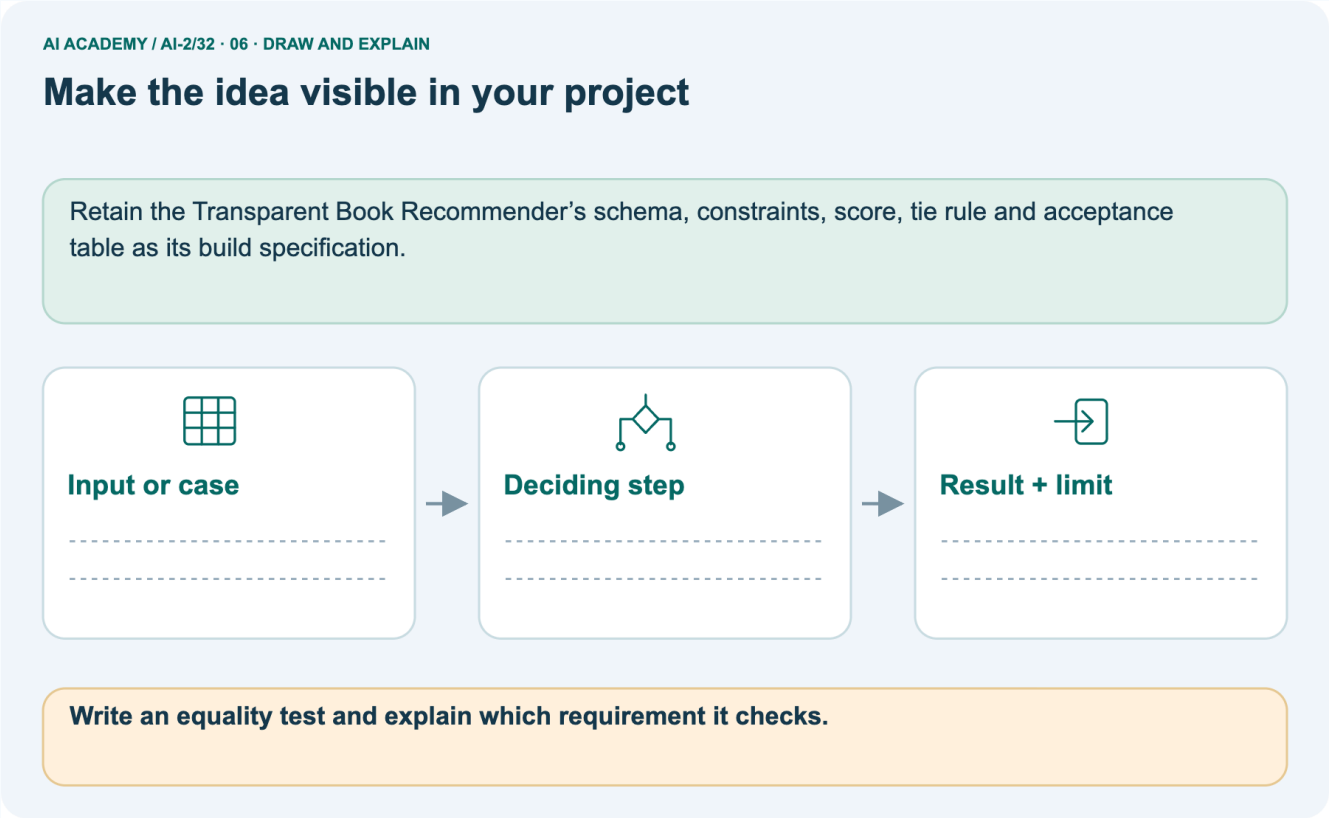
Planning fixes what success means before code returns an answer. The duration boundary includes equality, and the tie policy prevents accidental list-order dependence.

A complete **output** may include why other rows were excluded. Test expectations should come from the contract rather than copying the first program result.

Limit: The supplied interest scores are fictional; the specification does not establish genuine learner preferences.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Retain the Transparent Book Recommender's schema, constraints, score, tie rule and acceptance table as its build specification. Draw your **input**, deciding step, result and limit.

Supported question: Write an equality test and explain which **requirement** it checks.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Project Problem and Success Test: the case

02. Follow the deciding steps

03. Follow the evidence

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.